

- **Random Forest algorithm**
- **Variable importance and partial dependency**



Applied Geodata Science II

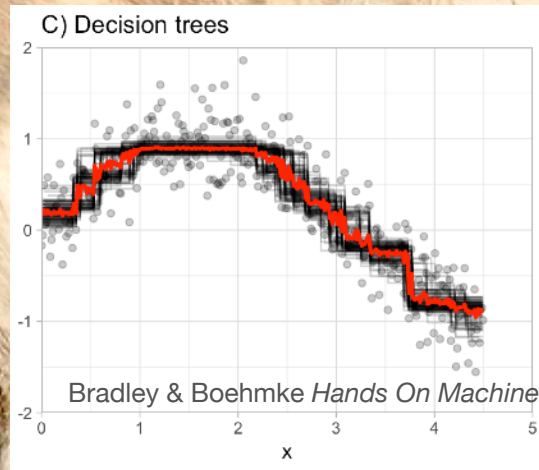
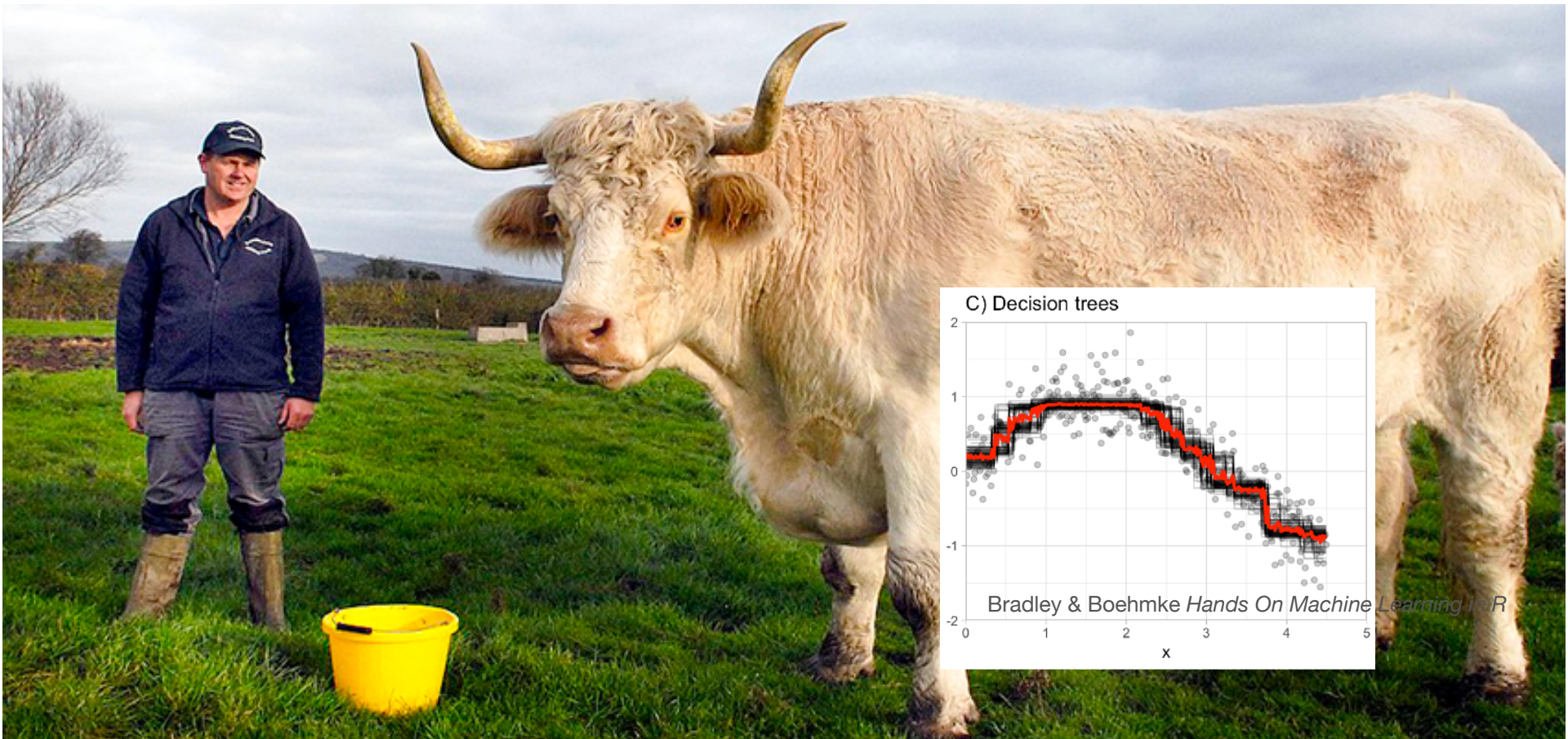
Session 4

Fabian Bernhard

07.10.2025

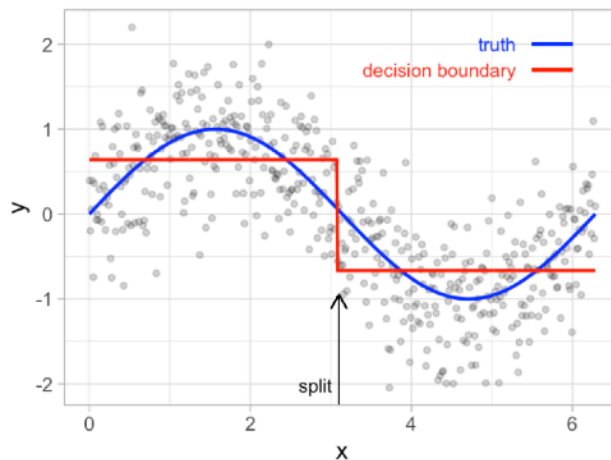
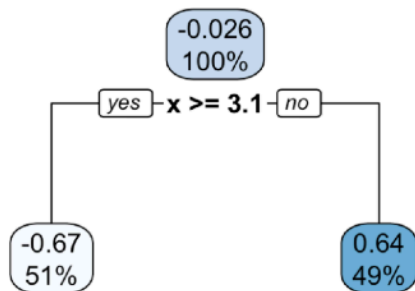


What's the weight of Field Marshall?

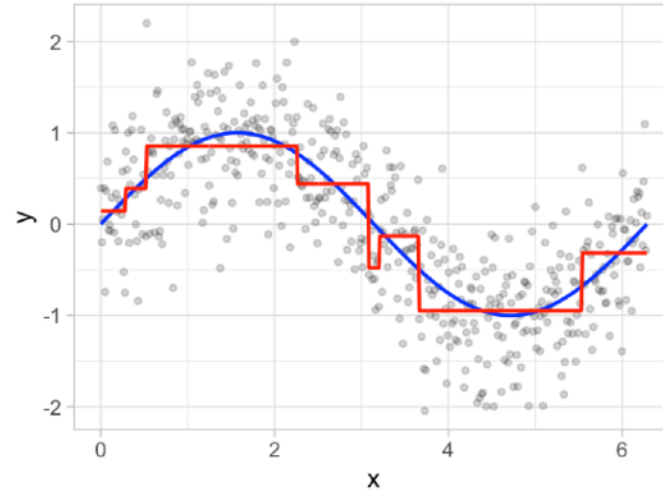
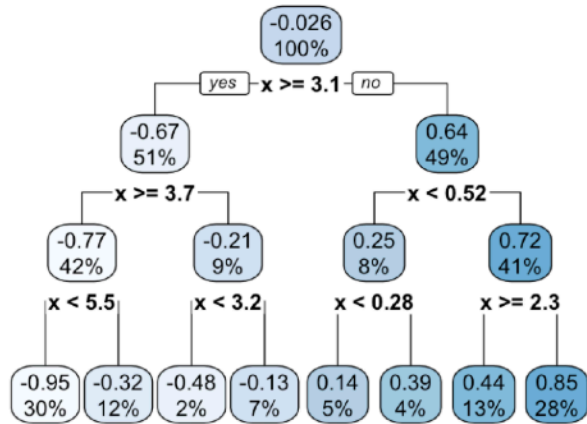


A decision tree

A decision tree



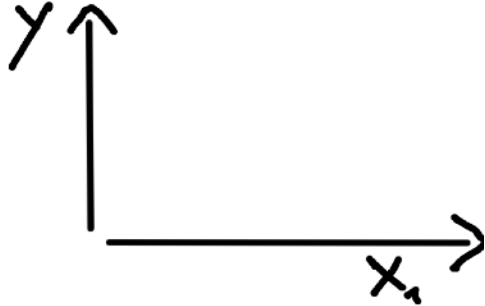
Tree growth



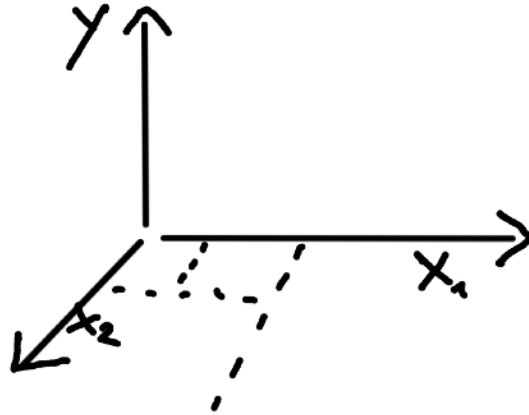
Bradley & Boehmke *Hands On Machine Learning in R*

More than one predictor

1 predictor



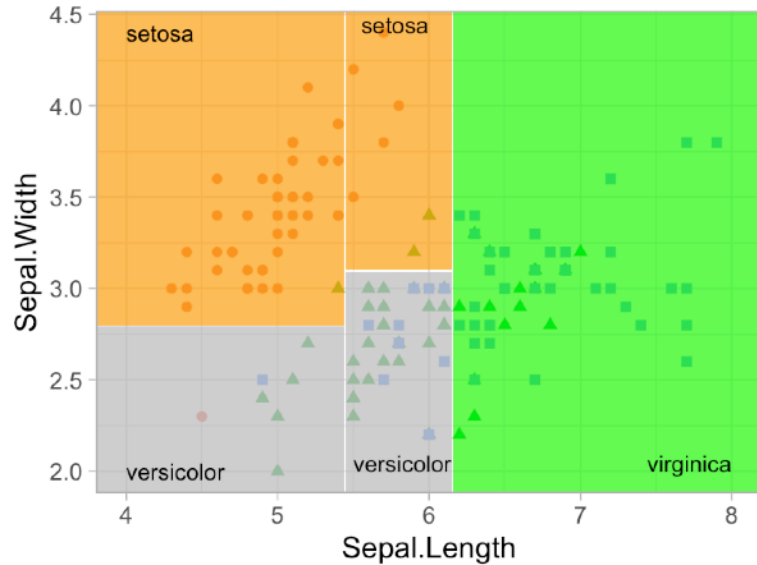
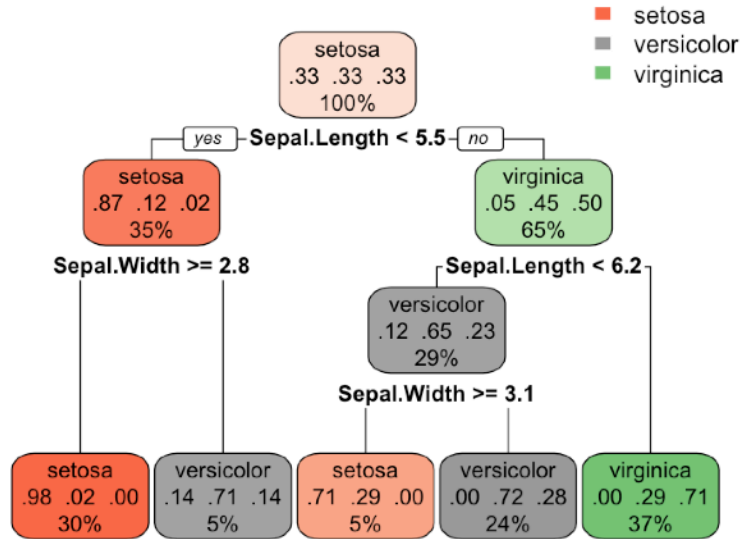
2 predictors



X predictors

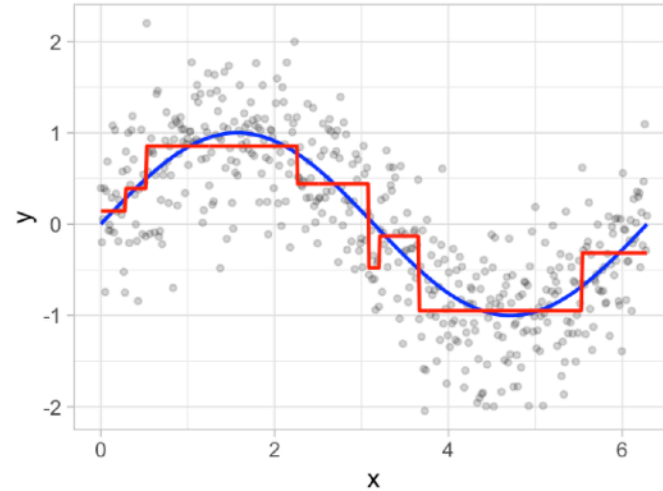
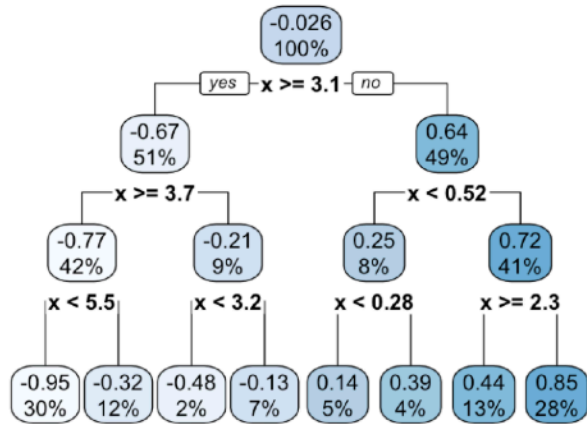
...

More than one predictor



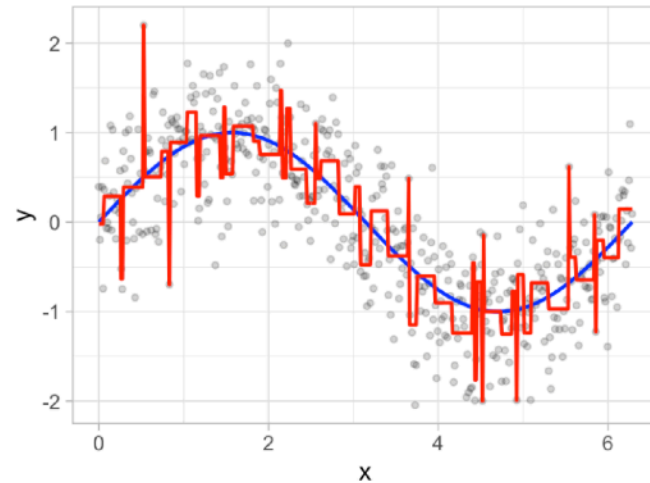
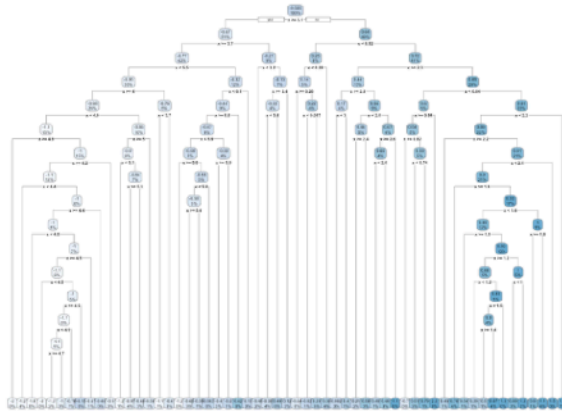
Bradley & Boehmke *Hands On Machine Learning in R*

Tree growth



Bradley & Boehmke *Hands On Machine Learning in R*

Too deep a tree (overfitted)



Bradley & Boehmke *Hands On Machine Learning in R*

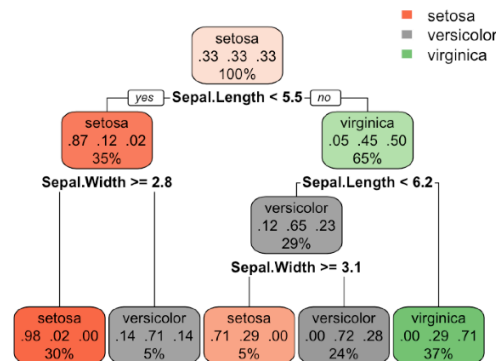
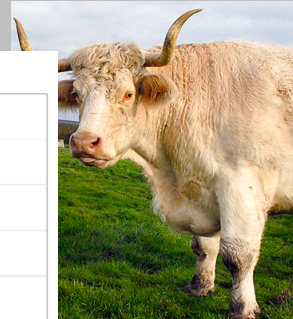
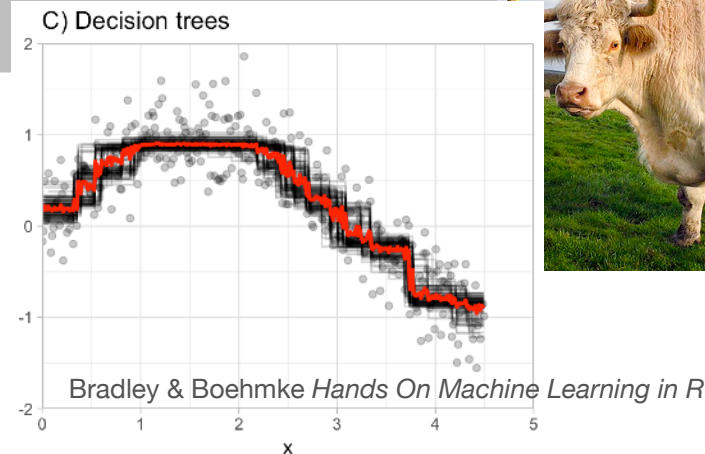
From a tree to a forest

For a robust estimate consider a forest of trees.

- number of trees

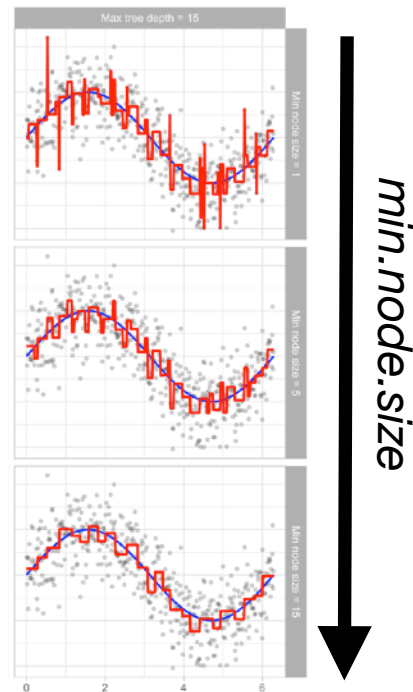
Averaging across all trees,
added randomness:

- data points for each tree
(bootstrap aggregation = bagging)
- predictor variables considered
for each decision (branch) within a tree



Hyperparameters in Random Forst

- *number of trees*
- *min.node.size*: the number of data points at the “bottom” of each decision tree (leaf nodes)
- *mtry*: number of predictors considered at each branch
- *splitrule*: the function applied for determining the goodness of a decision (default:
 - SSE for regression,
 - Gini impurity for classification)



Interpretable machine learning

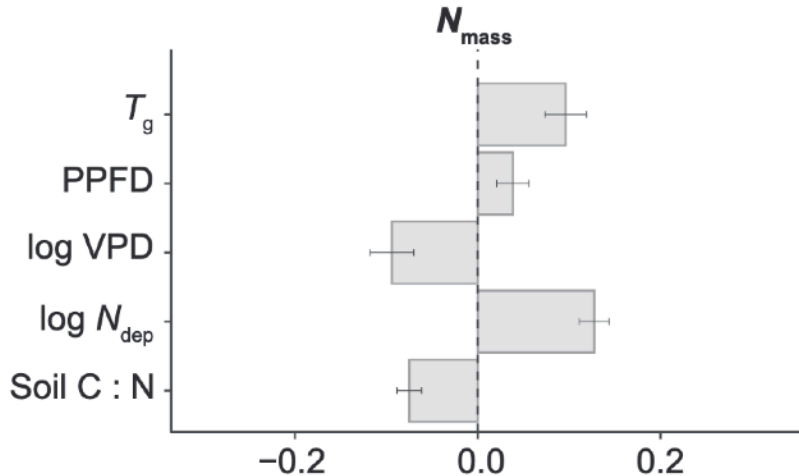
Shining light into the black box

Interpreting coefficients of a linear model

```
linmod_nmass <- lm(  
  log_Nmass ~ scale(log_vpd) + scale(ppfd) + scale(tgrowth) + scale(CNrt) + scale(log_ndep),  
  data = df  
)
```

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots \beta_N x_N$$

$$Y = \beta_0 + \sum_{i=1}^N \beta_i x_i$$



```
> summary(linmod_nmass)
```

Call:

```
lm(formula = log_Nmass ~ scale(log_vpd) + scale(ppfd) + scale(tgrowth) +  
  scale(CNrt) + scale(log_ndep), data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.48710	-0.27452	0.00131	0.26255	1.41388

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.033257	0.007085	-569.259	< 2e-16 ***
scale(log_vpd)	-0.093902	0.012500	-7.512	7.54e-14 ***
scale(ppfd)	0.038489	0.009249	4.161	3.25e-05 ***
scale(tgrowth)	0.096150	0.011864	8.104	7.53e-16 ***
scale(CNrt)	-0.074868	0.007193	-10.409	< 2e-16 ***
scale(log_ndep)	0.127458	0.008647	14.741	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3966 on 3128 degrees of freedom
(9 observations deleted due to missingness)

Multiple R-squared: 0.0966, Adjusted R-squared: 0.09515

F-statistic: 66.89 on 5 and 3128 DF, p-value: < 2.2e-16

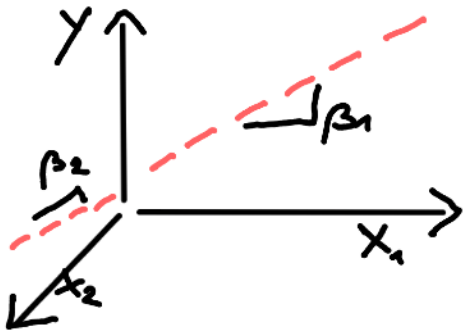
```
> coef(linmod_nmass)
```

(Intercept)	scale(log_vpd)	scale(ppfd)	scale(tgrowth)	scale(CNrt)	scale(log_ndep)
-4.03325690	-0.09390211	0.03848865	0.09614960	-0.07486796	0.12745766

How to interpret Random Forests?

Linear regression

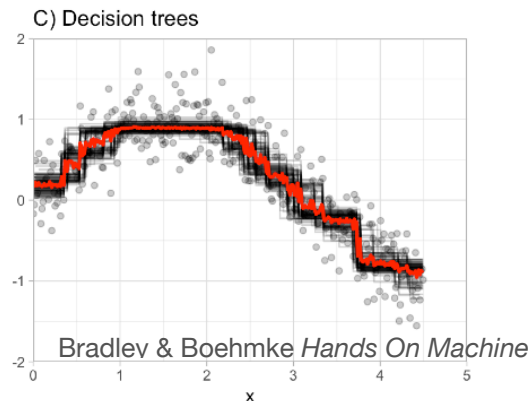
$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots \beta_N x_N$$



Use β !

versus

Random forest (or other ML alg.)



... no formula to analyse.

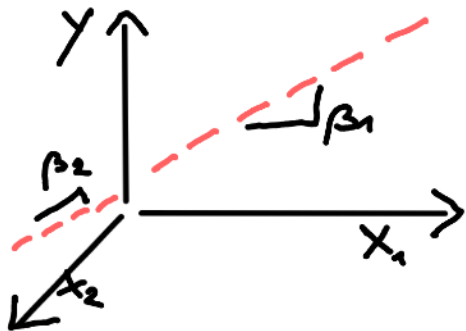
How to interpret Random Forests?

Linear regression

versus

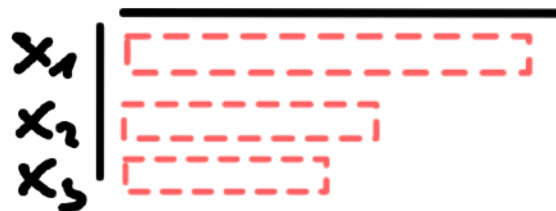
Random forest (or other ML alg.)

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots \beta_N x_N$$

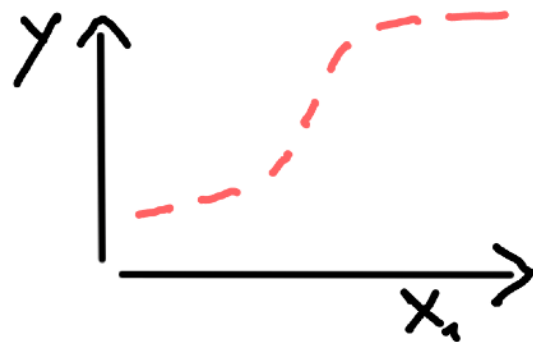


Use β !

Variable Importance:



Partial Dependency plot:



Variable importance

- Measures and compares how strongly the model skill depends on the information contained in the data of each variable.

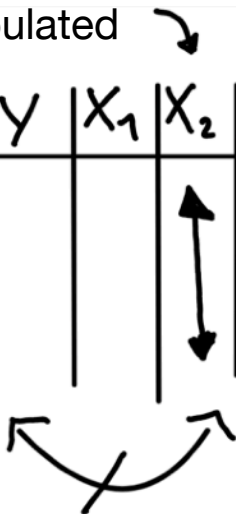
1. Compute loss function L for model trained on un-manipulated data
2. For predictor variable i in $\{1, \dots, p\}$ do
 - | Permute values of variable i .
 - | Fit model.
 - | Estimate loss function L_i .
 - | Compute variable importance as $I_i = L_i/L$ or $I_i = L_i - L_0$.
3. Sort variables by descending values of I_i .

original

Y	X_1	X_2	X_3

manipulated

Y	X_1	X_2	X_3

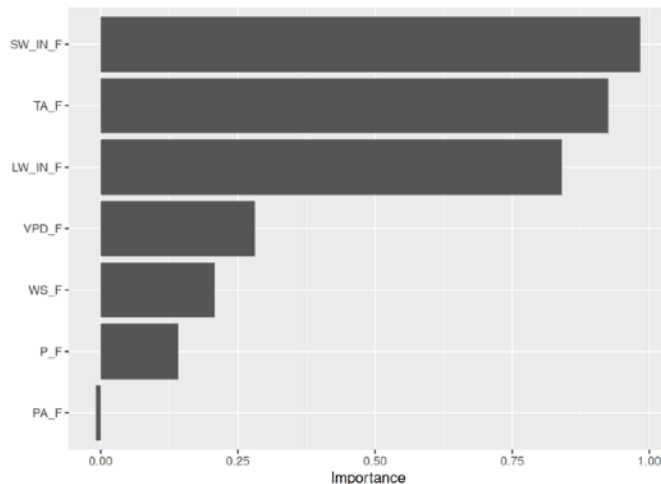


Variable importance

- Measures and compares how strongly the model skill depends on the information contained in the data of each variable.

```
vip::vip(rf_mod,                # Fitted model object
  train = rf_mod$trainingData |> # Training data used in the model
  dplyr::select(-TIMESTAMP),
  method = "permute",           # VIP method
  target = "GPP_NT_VUT_REF",    # Target variable
  nsim = 5,                     # Number of simulations
  metric = "RMSE",              # Metric to assess quantify permutation
  sample_frac = 0.75,           # Fraction of training data to use
  pred_wrapper = predict        # Prediction function to use
)
```

1. Compute loss function L for model trained on un-manipulated data
2. For predictor variable i in $\{1, \dots, p\}$ do
 - | Permute values of variable i .
 - | Fit model.
 - | Estimate loss function L_i .
 - | Compute variable importance as $I_i = L_i/L$ or $I_i = L_i - L_0$.End
3. Sort variables by descending values of I_i .



Partial dependence plots

- Illustrates how the target variable varies as a function of one predictor, while all the remaining predictors (co-vary) as observed in the data used for model training.

original

y	X_1	X_2	X_3

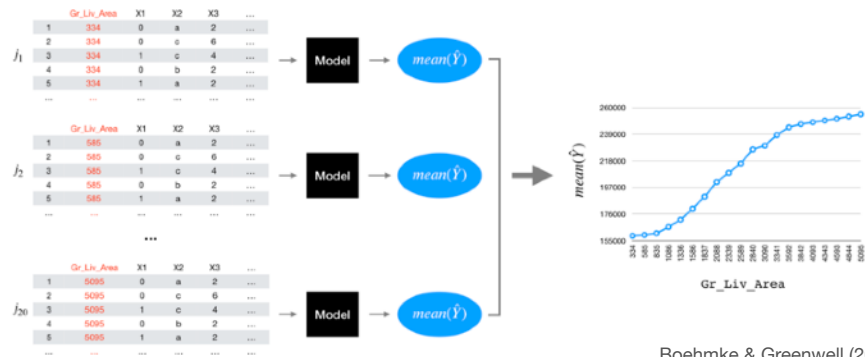
manipulated ↘

y	X_1	X_2	X_3



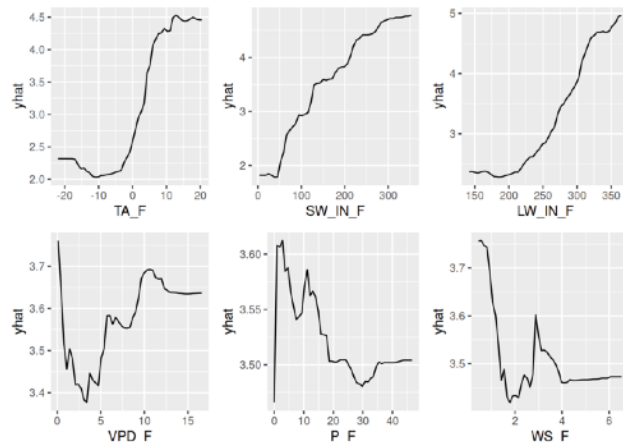
Partial dependence plots

- Illustrates how the target variable varies as a function of one predictor, while all the remaining predictors (co-vary) as observed in the data used for model training.



Boehmke & Greenwell (2019)

```
# The partial() function can take n=3 predictors at max and will try to create
# a n-dimensional visualisation to show interactive effects. However,
# this is computational intensive, so we only look at the simple
# response-predictor plots
all_plots <- purrr::map(
  preds,
  ~pdp::partial(
    rf_mod,      # Model to use
    .,           # Predictor to assess
    plot = TRUE, # Whether output should be a plot or dataframe
    plot.engine = "ggplot2" # to return ggplot objects
  )
)
```



git - some difficulties experienced

- [] use ssh link instead of https check with `git remote -v`
- [] no tick gpg
- [] work in OneDrive (make available offline)
- [] `git pull`, `git commit`, `git push`

Variable importance in action

nature communications



Article

<https://doi.org/10.1038/s41467-024-49665-4>

Environmental versus phylogenetic controls on leaf nitrogen and phosphorous concentrations in vascular plants

Received: 12 September 2023

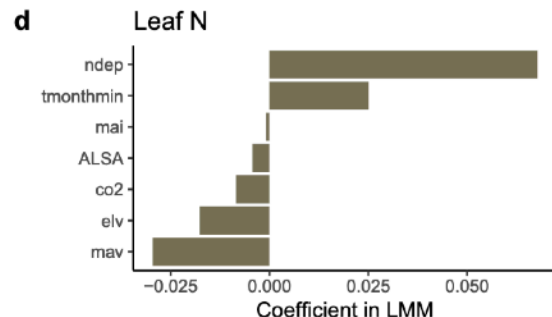
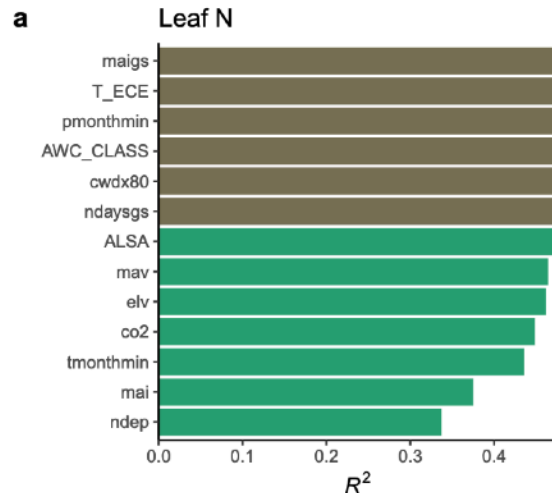
Accepted: 15 June 2024

Published online: 24 June 2024

Check for updates

Di Tian^{1,2,3} , Zhengbing Yan⁴, Bernhard Schmid^{5,6}, Jens Kattge^{7,8}, Jingyun Fang⁶ & Benjamin D. Stocker^{2,3,9,10}

Global patterns of leaf nitrogen (N) and phosphorus (P) stoichiometry have been interpreted as reflecting phenotypic plasticity in response to the environment, as well as overriding effect of the distribution of species growing in their



Partial dependence plots in action

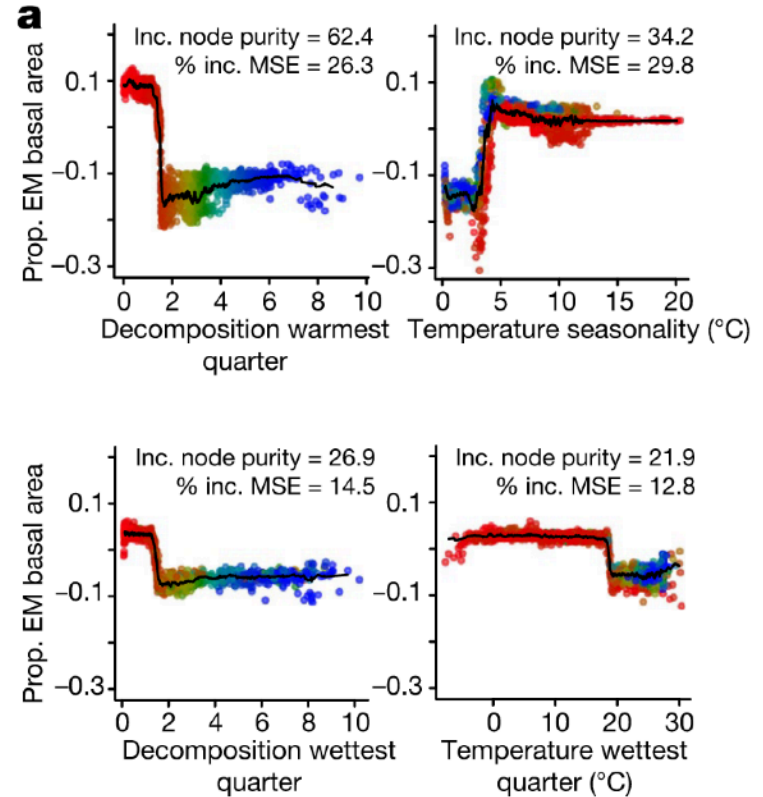
LETTER

Corrected: Author Correction

<https://doi.org/10.1038/s41586-019-1128-0>

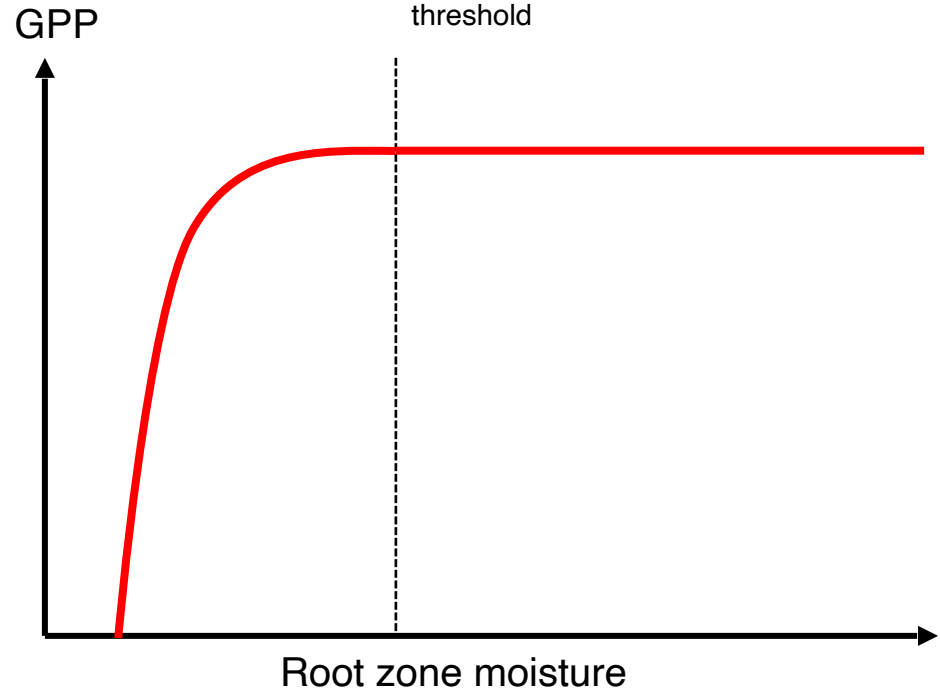
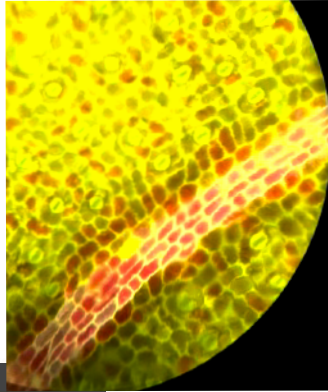
Climatic controls of decomposition drive the global biogeography of forest–tree symbioses

B. S. Steidinger^{1,15}, T. W. Crowther^{2,15*}, J. Liang^{3,4,15*}, M. E. Van Nuland¹, G. D. A. Werner⁵, P. B. Reich^{6,7}, G. J. Nabuurs^{8,162}, S. de-Miguel^{9,10}, M. Zhou³, N. Picard¹¹, B. Herault^{12,13}, X. Zhao⁴, C. Zhang⁴, D. Routh², GFBI consortium¹⁴ & K. G. Peay^{1*}



Partial dependence plots in action

Understanding root zone moisture limitation on GPP



Diagnose partial dependency?

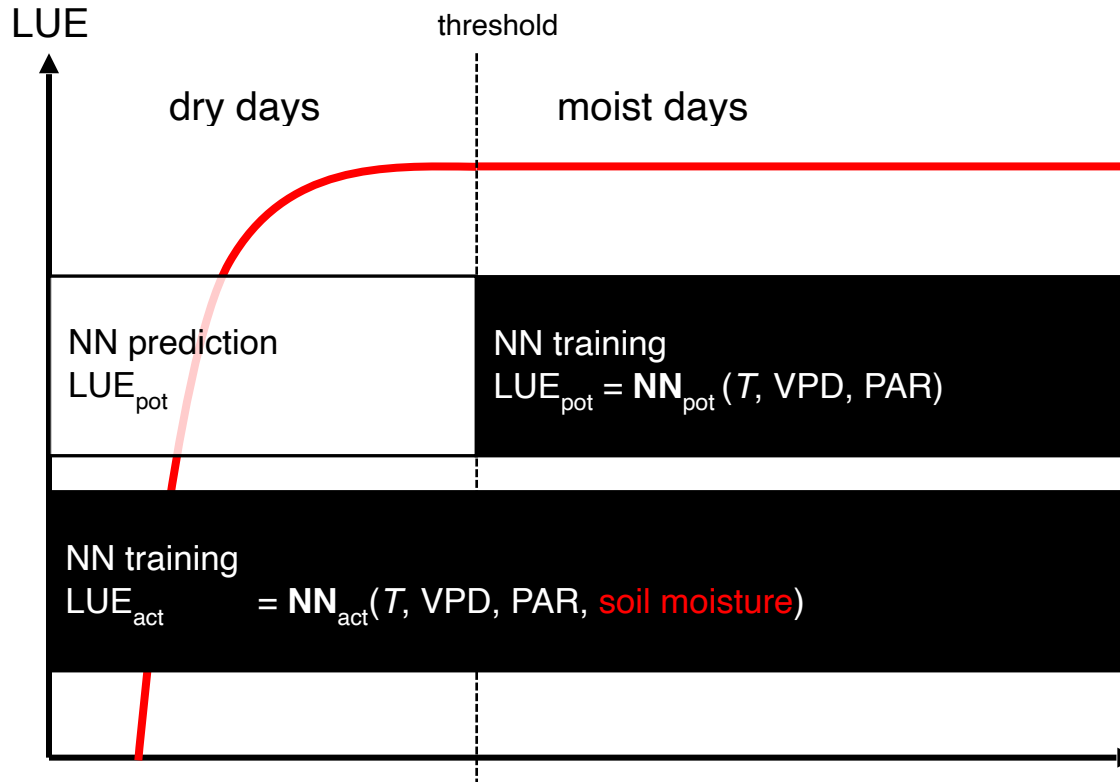


Understanding water limitation on GPP

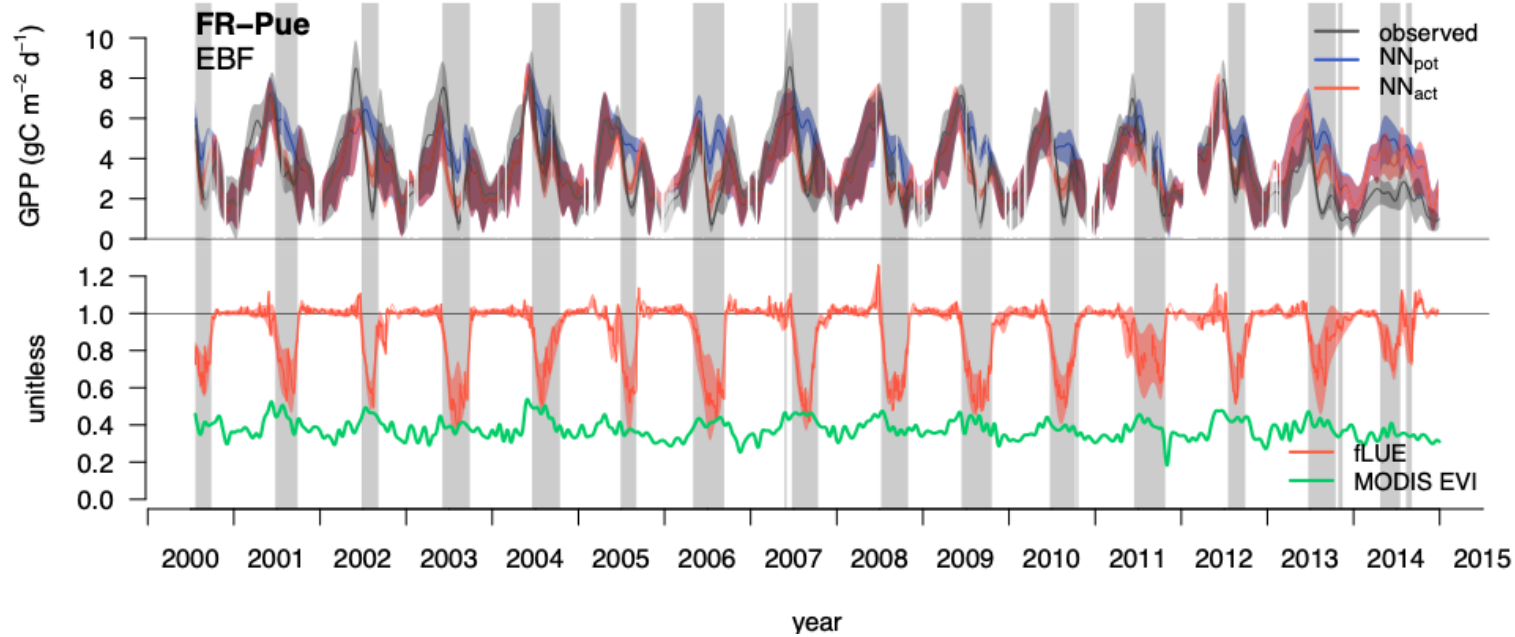
$GPP = (\text{vegetation greenness}) \times \text{light} \times (\text{light use efficiency})$

➡ $LUE = GPP / (\text{vegetation greenness} \times \text{light})$

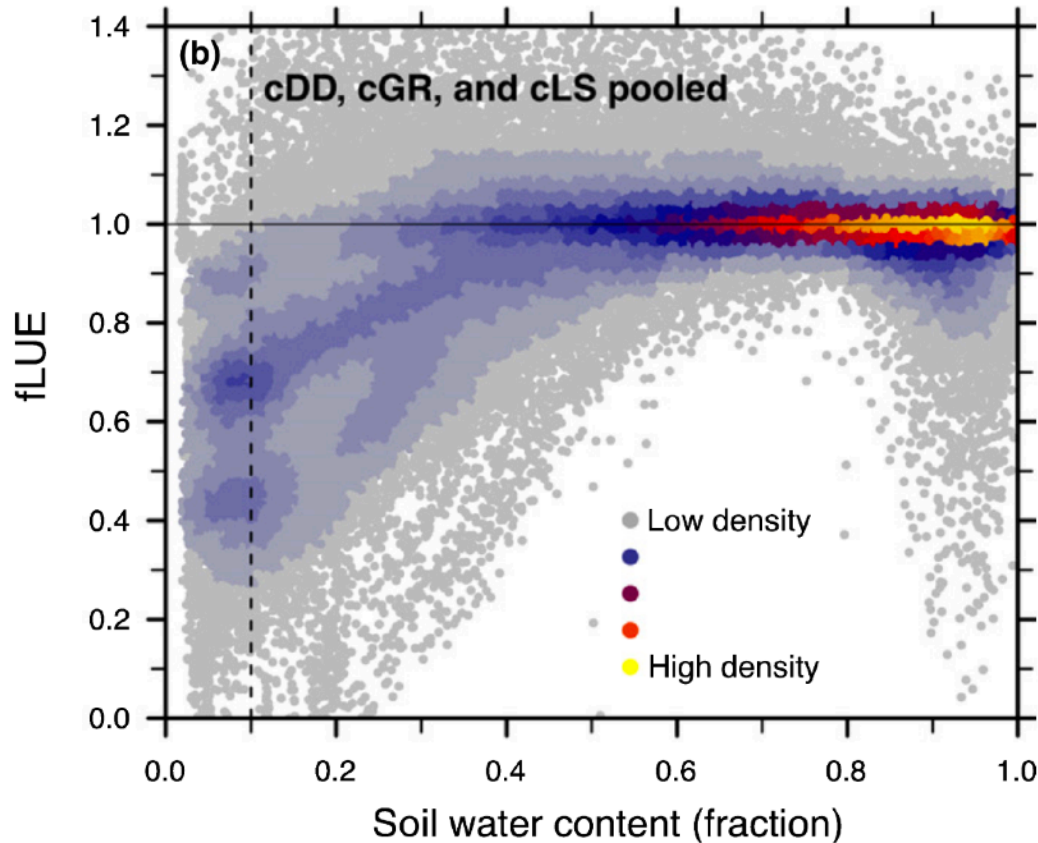
➡ $fLUE = LUE_{act} / LUE_{pot}$



Soil moisture effect separated



Partial dependency



- Diversity of responses across sites
- Unreliable soil moisture data